

ESP32 code

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/******
```

Smart Class++room with Blynk IoT

ESP32 + DHT22 + PIR Sensors

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```

```
//===== BLYNK =====
```

```
#define BLYNK_TEMPLATE_ID "TMPL3igoJN-lj"
```

```
#define BLYNK_TEMPLATE_NAME "Smart Zonal Energy optimization"
```

```
#define BLYNK_AUTH_TOKEN "o4CfaWvBNIV7nsMR-jVI2GbKEpJOz_Dn"
```

```
//===== Libraries =====
```

```
#include <WiFi.h>
```

```
#include <BlynkSimpleEsp32.h>
```

```
#include "DHT.h"
```

```
//===== WiFi =====
```

```
char ssid[] = "Wokwi-GUEST";
```

```
char pass[] = "";
```

```
//===== Pins =====
```

```
#define PIR1 14
```

```
#define PIR2 27
```

```
#define LIGHT1 18
```

```
#define FAN1 19
```

```
#define LIGHT2 23
```

```
#define FAN2 25
```

```
#define ENTRY_BUTTON 32
```

```
#define EXIT_BUTTON 34
```

```
#define DHTPIN 4
```

```

#define DHTTYPE DHT22

#define VENTILATION_LED 26

//===== Objects =====

DHT dht(DHTPIN, DHTTYPE);

BlynkTimer timer;

//===== Variables =====

int students = 0;

bool zone1Active = false;

bool zone2Active = false;

bool ventilationActive = false;

float temperature = 24.0;

bool prevEntryState = HIGH;

bool prevExitState = HIGH;

//===== Energy =====

const int lightPower = 15; // Watts

const int fanPower = 60; // Watts

const int ventPower = 40; // Watts

const float supplyVoltage = 230.0;

//=====

void sendToBlynk()

{

int totalPower = 0;

if (zone1Active)

totalPower += lightPower + fanPower;

if (zone2Active)

totalPower += lightPower + fanPower;

if (ventilationActive)

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totalPower += ventPower;

float current = (float)totalPower / supplyVoltage;

// Send values to Blynk

Blynk.virtualWrite(V0, students);

Blynk.virtualWrite(V1, temperature);

Blynk.virtualWrite(V2, zone1Active ? 1 : 0);

Blynk.virtualWrite(V3, zone2Active ? 1 : 0);

Blynk.virtualWrite(V5, supplyVoltage);

Blynk.virtualWrite(V6, current);

Blynk.virtualWrite(V7, totalPower);

// Serial Monitor

Serial.println("=====");

Serial.print("Students : ");

Serial.println(students);

Serial.print("Temperature : ");

Serial.print(temperature);

Serial.println(" C");

Serial.print("Zone 1 : ");

Serial.println(zone1Active ? "ON" : "OFF");

Serial.print("Zone 2 : ");

Serial.println(zone2Active ? "ON" : "OFF");

Serial.print("Voltage : ");

Serial.print(supplyVoltage);

Serial.println(" V");

Serial.print("Current : ");

Serial.print(current, 2);

Serial.println(" A");
```

```

Serial.print("Power : ");

Serial.print(totalPower);

Serial.println(" W");

Serial.println("=====");

}

//=====

void setup()

{

Serial.begin(115200);

pinMode(PIR1, INPUT);

pinMode(PIR2, INPUT);

pinMode(LIGHT1, OUTPUT);

pinMode(FAN1, OUTPUT);

pinMode(LIGHT2, OUTPUT);

pinMode(FAN2, OUTPUT);

pinMode(ENTRY_BUTTON, INPUT_PULLUP);

pinMode(EXIT_BUTTON, INPUT_PULLUP);

pinMode(VENTILATION_LED, OUTPUT);

dht.begin();

Serial.println("Connecting to WiFi and Blynk...");

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

Serial.println("Connected!");

timer.setInterval(1000L, sendToBlynk);

}

//=====

void loop()

{

```

```
Blynk.run();

timer.run();

// Entry Button

bool entryState = digitalRead(ENTRY_BUTTON);

if (entryState == LOW && prevEntryState == HIGH)

{

students++;

Serial.println("Student Entered");

delay(200);

}

prevEntryState = entryState;

// Exit Button

bool exitState = digitalRead(EXIT_BUTTON);

if (exitState == LOW && prevExitState == HIGH)

{

if (students > 0)

students--;

Serial.println("Student Exited");

delay(200);

}

prevExitState = exitState;

// Read Sensors

int motion1 = digitalRead(PIR1);

int motion2 = digitalRead(PIR2);

float temp = dht.readTemperature();

if (!isnan(temp))

temperature = temp;
```

```
// Reset outputs

zone1Active = false;

zone2Active = false;

ventilationActive = false;

if (students > 0)

{

    if (motion1 == HIGH)

        zone1Active = true;

    if (motion2 == HIGH)

        zone2Active = true;

    if (zone1Active || zone2Active)

        ventilationActive = true;

}

// Outputs

digitalWrite(LIGHT1, zone1Active);

digitalWrite(FAN1, zone1Active);

digitalWrite(LIGHT2, zone2Active);

digitalWrite(FAN2, zone2Active);

digitalWrite(VENTILATION_LED, ventilationActive);

}
```